

SWYAM VATWANI

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SUMMARY

Computer Science undergraduate at NIT Delhi with experience building end-to-end AI and full-stack applications using Python, FastAPI, Next.js, MERN stack, Docker, and machine learning. Skilled in DSA, REST APIs, ML model development, recommendation systems, SHAP explainability, and RAG-based applications. Passionate about building scalable software and intelligent systems that solve real-world problems.

EDUCATION

National Institute of Technology Delhi

2024 – 2028

Bachelor of Technology, Computer Science Engineering - GPA – 9.0

New Delhi, India

EXPERIENCE

Freelance Web Developer [↗](#) French Language Learning Platform | Freelance | 12/2025 - 2/2026

- Developed and deployed a production-ready language learning platform using the **MERN** stack, Sanity **CMS**, and Mailchimp for marketing automation. Configured and deployed the application on **Vercel** with custom domain integration through **GoDaddy**.
- Integrated **analytics and monitoring** tools to track user behavior and website performance.
- Achieved **230+** website visitors after launch.
- Enhanced search engine visibility through **SEO** best practices, sitemap generation, and optimized metadata.

PROJECTS

AI Career & Salary Intelligence Platform [↗](#)

06/2025

Next.js, FastAPI, Docker, Hugging Face Spaces, Vercel, scikit-learn, SHAP, Sentence-Transformers, Gemini API

- Built and deployed a **full-stack ML platform** that analyzes resumes, recommends jobs, predicts salary, explains predictions, and generates career roadmaps.
- Trained salary prediction models on a **250k-row dataset**, achieving $R^2 = 0.963$, $MAE \approx ₹5.4k$, and $RMSE \approx ₹7.1k$ with Linear Regression.
- Linear Regression outperformed Gradient Boosting by **16.7% lower MAE** and **14.1% lower RMSE** on the sampled benchmark.
- Implemented SHAP explanations to show feature-level salary impact and built a job recommendation engine over **75k job records** using **sentence transformers**, **TF-IDF** and **cosine similarity**.
- Integrated **Gemini API** for roadmap generation with a **rule-based fallback** and deployed the app using **Docker**, **Hugging Face Spaces**, **Vercel**, and **Git LFS**.

StudyGPT | AI-Powered Learning Platform [↗](#)

04/2025

RAG, MongoDB Atlas Vector Search, Gemini Embeddings, LLM Reranking, LLM-as-a-Judge

- Developed an AI-powered learning platform using **Retrieval-Augmented Generation (RAG)**, **MongoDB Atlas Vector Search**, **Gemini Embeddings**, **LLM-based reranking**, and a **multi-model AI pipeline**.
- Implemented automated flashcard generation, quiz generation, summarization, and concept explanation features to enhance learning outcomes.
- Designed a retrieval pipeline combining **vector search**, embedding-based **semantic search**, and reranking to improve response relevance and accuracy.
- Integrated LLM-as-a-Judge evaluation to assess and validate AI-generated responses for quality and reliability.

ACHIEVEMENTS

- Patent on A System and Method for Energy Efficient Redirection and Distribution of Light
- Solved **300+** DSA problems covering arrays, trees, graphs, DP, recursion, and greedy algorithms.
- JEE ADVANCED, 2022** - Secured All India Rank of 4576 in general category (among 1,80,000 candidates).
- JEE MAINS, 2024** - Secured All India Rank of 5885 in general category (among 14,76,557 candidates).

TECHNICAL SKILLS

- Languages & CS:** Python, Java, C, C++, JavaScript, SQL, Data Structures, Algorithms, OOP, DBMS, OS
- ML/AI:** Machine Learning, Deep Learning, NLP, Regression, SHAP, RAG, Sentence-Transformers, Model Evaluation
- Frameworks/Libraries:** scikit-learn, TensorFlow, Pandas, NumPy, FastAPI, Flask
- Full Stack/Tools:** React.js, Next.js, Node.js, Express.js, MongoDB, Tailwind CSS, Docker, Git, GitHub